

Virtual Mouse Using Hand Gestures

A Synopsis

for

Mini Project

**BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE &
ENGINEERING**

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Introduction

The evolution of Human-Computer Interaction has reached a stage where physical touch is no longer the only medium for digital communication. The "Virtual Mouse Using Hand Gestures" is an AI-driven system designed to replace traditional, mechanical hardware peripherals with intuitive, touchless hand movements.

By utilizing advanced Computer Vision libraries like OpenCV and MediaPipe, the system captures a real-time webcam feed to continuously track 21 distinct hand landmarks. It instantly translates the geometric relationships between these fingertips—such as the distance between the index finger and thumb—into precise screen coordinates and OS-level mouse commands, including navigation, clicking, and scrolling.

This frictionless approach is particularly valuable for advancing spatial computing and providing sanitary alternatives in sterile environments where touching physical hardware poses a risk. Furthermore, it significantly enhances digital accessibility for users with physical limitations, offering a seamless, hardware-independent computing experience.

Literature Review

Traditional input devices like the optical mouse have existed for decades but carry ergonomic and environmental limitations [1]. Recent studies have demonstrated the power of MediaPipe, a Google-developed framework capable of tracking 21 3D landmarks on a human hand with high precision [2]. Unlike previous methods that required depth-sensing cameras or infrared sensors, current research focuses on using standard RGB webcams to process frames through Convolutional Neural Networks (CNNs), making the technology accessible on standard consumer hardware [3], [4].

Problem Definition

The reliance on physical mice presents several recurring issues:

- **Hardware Fatigue:** Mechanical parts like buttons and scrolls eventually fail, leading to electronic waste.
- **Space Constraints:** Physical mice require a flat, consistent surface to operate effectively.
- **Ergonomics:** "Mouse Shoulder" and Carpal Tunnel syndrome are common among long-term computer users.
- **Sanitation:** In public healthcare or industrial settings, shared hardware can lead to the transmission of bacteria.

Objectives

- To develop a real-time tracking system using a standard laptop webcam.
- To accurately map hand landmarks to screen coordinates using mathematical interpolation.
- To implement a complete set of mouse functions: Cursor Movement, Left Click, Right Click, and Scrolling.
- To optimize the system for low-latency performance on modern processor architectures.

Methodology

The proposed system utilizes a **modular, function-based architecture** in Python, relying on standalone functional blocks executed in a continuous real-time loop. The pipeline follows these key steps:

1. **Input Stream:** A capture function reads live webcam video at 30 FPS using OpenCV.
2. **Detection:** MediaPipe functions process each frame to detect the hand and extract the exact (x, y) coordinates of the fingertips.
3. **Coordinate Scaling:** The `numpy.interp` function mathematically scales the coordinates from the camera resolution to the screen resolution.
4. **Gesture Mapping Logic:**
 - **Movement:** A tracking function uses the index fingertip coordinates to move the pointer.
 - **Click Actions:** A mathematical function calculates the Euclidean distance between the index and thumb. If the distance drops below a threshold, it registers a "click."
5. **Execution:** Finally, `PyAutoGUI` functions execute the translated commands (cursor movement, clicks, scrolling) directly on the Operating System.

References

Books References

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